Sensing & Perception

Lab 2 – Signal conditioning using the strain gauge load cells

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1. Introduction

In any Data Acquisition System (DAS), signals obtained from sensors are often not directly usable. Signal conditioning refers to the process of manipulating sensor outputs to prepare them for digitization and further processing. As shown in Figure 1, signal conditioning is one of the key steps in the data acquisition process.

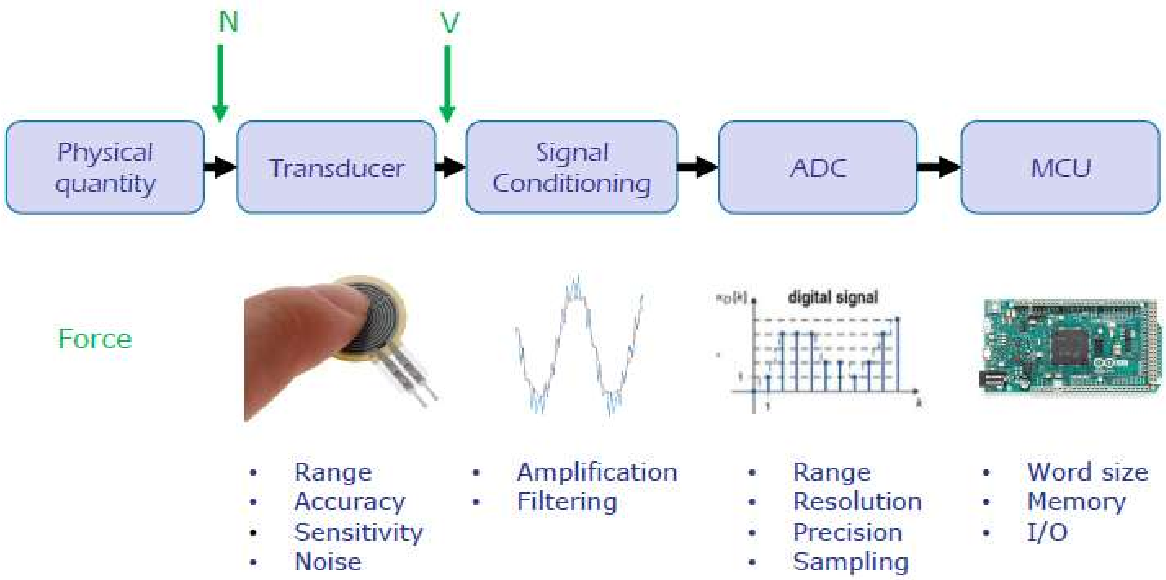

Figure 1. Schematic representation of the processing stages of a Data Acquisition System

The objective of this lab is to understand the different steps of the signal conditioning by using the straight bar load cell TAL220B strain gauge (as represented in Figure 2).

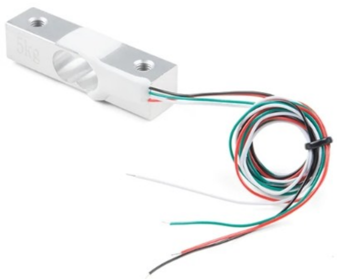

Figure 2. Load Cell - 5kg, Straight Bar (TAL220B)

The maximum load supported by the strain gauge is 5 kg. Therefore, it is important to use only masses below 5 kg during the experiment. <u>It is strictly forbidden to apply any deformation by hand, as it may. damage the equipment.</u>

The straight bar load cell TAL220B strain gauge can measure the electrical resistance that changes in response to the applied strain. These load cells have four strain gauges that are hooked up in a Wheatstone bridge formation (Figure 3). When the bridge is perfectly balanced (all resistances are equal or symmetrically varied), the voltage difference V between the midpoints is zero. Under strain, the resistances of the gauges change differently, unbalancing the bridge and producing a measurable output voltage proportional to the applied strain.

This method enables highly accurate and sensitive strain measurements. Depending on the direction of the applied force, it is possible to obtain positive or negative voltage values from the TAL220B load cell.

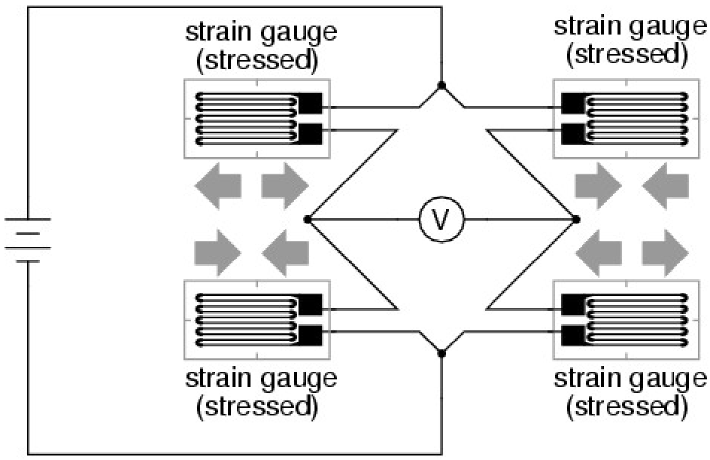

Figure 3. Full bridge strain gauge circuit

The different steps of the lab are:

- Analysis of data obtained under mechanical loading
- Amplification of low-level signals using an instrumentation amplifier
- Signal shifting and scaling to match a 0 - 5 V ADC input range
- Comparison of the strain gauge load cells, your homemade sensor (Lab1) and the flexiforce sensor (Lab1)

Required equipment for the lab:

- ±12 V Power supply
- 1 table multimeter fluke 45
- 1 straight bar load cell TAL220B strain gauge 5 kg
- 1 instrumentation amplifier INA826
- 1 operational amplifier LM741
- 1 breadboard
- 100 kΩ resistor (×2)
- 470 kΩ potentiometer (×1)
- 100 nF capacitor (×4)
- 100 Ω resistor (×1)

2. Calibration

The first part of the lab is intended to understand the data obtained under mechanical loading.

- To achieve this first part, apply a voltage of 0 - 12 V to the straight bar load cell TAL220B strain gauge. The strain gauge wiring is shown below (Figure 4). The supply voltage should be connected to the "*Exc connections*" and the output voltage should be measured from the "*Sig connections*".

Please have the teacher check your circuit before switching on the voltage source.

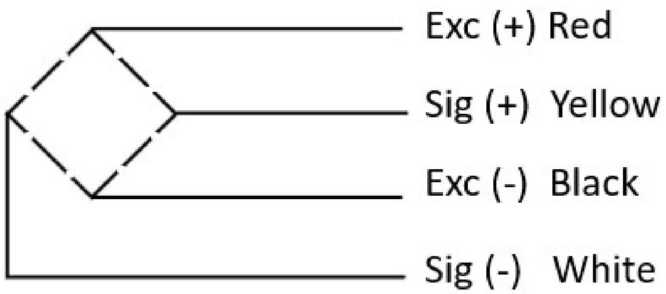

Figure 4. Wiring of the used straight bar load cell strain gauge

- Measure the output voltage of the circuit with no strain applied.
- Place different masses on the strain gauge and measure the output voltage corresponding to the mechanical stress.

  Perform at least two additional series of measurement to estimate the repeatability of the sensor. Conclude on the repeatability of the strain gauge.

  Is the sensitivity of the system linear?

3. Amplification of the output signal

The output values are too small to be used. To process the sensor's data effectively, we will insert an amplification circuit to amplify the strain gauge output values. The instrumentation amplifier INA826 (Figure 5) will be used to get an amplification factor of ≈ 500.

Look for information about the instrumentation amplifier and give the advantages of using an instrumentation amplifier in this application.

The relation between the output signal and the input signal of the instrumentation amplifier is given by: $V_o = G \times (V_{IN+} - V_{IN-}) + REF$ with $G = 1 + \frac{2\times 24.7\ kohm}{R_G}$

The gain is set by a single external resistor R_G connected between pins 2 and 3. Refer to figure 5 and 6 for the location of pins 2 and 3.

The output of the INA826 is referred to the output reference (REF) terminal, which is normally grounded. Sometimes, adjustments can be made by applying a voltage to the REF terminal. The voltage applied to the REF terminal is summed at the output value.

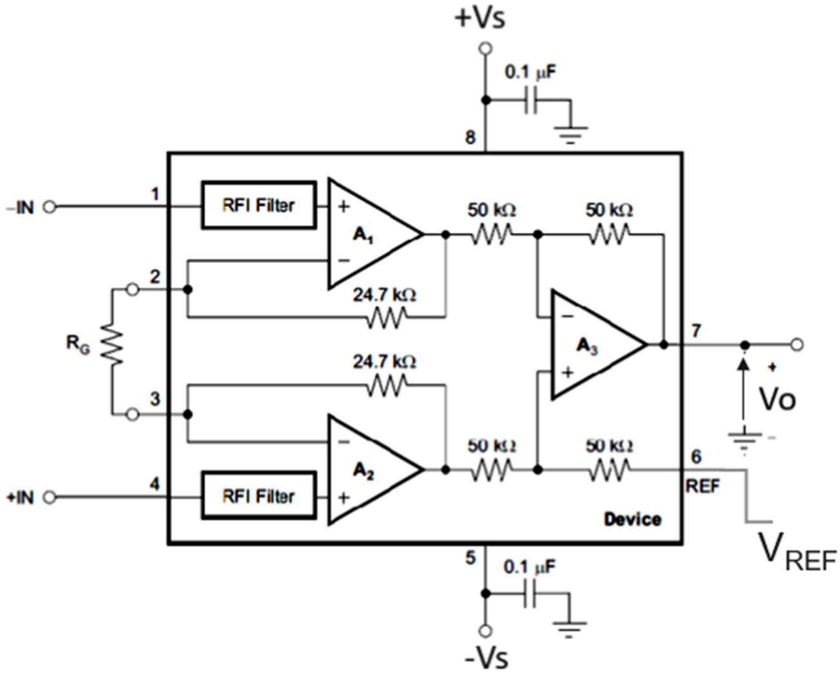

Figure 5. Functional block diagram of the instrumentation amplifier

- Calculate the required value of the resistance R_G to achieve an amplification factor G of 500.

The pin configuration of the instrumentation amplifier is represented in Figure 6. You can find more information about the instrumentation amplifier in file "datasheet_INA826". For this lab, the 1st pin is marked with the white colour.

Select a resistor with a value close to the calculated resistance and connect it between pin 2 and pin 3 of the instrumentation amplifier.

Connect the output of the strain gauge to the inputs (-IN and +IN) of the instrumentation amplifier INA826 and supply the instrumentation amplifier with ± 12 V ($+V_S$=12 V and $-V_S$= -12 V). For this first part, the REF terminal of the instrumentation amplifier should be connected to ground.

Please have the teacher check your circuit before switching on the voltage source.

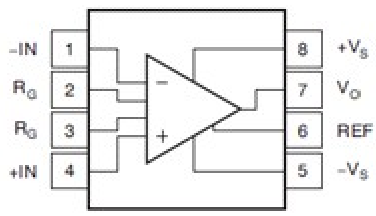

Figure 6. Pin configuration of the instrumentation amplifier

Place the different masses on the strain gauge and measure the output voltage Vo.
Check the amplification factor of the instrumentation amplifier.

4. Signal shifting and scaling of the amplified output values

Depending on the direction of the applied force on the strain gauge, the output signal can become negative. In some Data acquisition system, it's better to have only positive values for single supply ADC. Therefore, it is necessary to shift the amplified signal to ensure it remains entirely within the positive voltage range.
To accomplish this level shift, a voltage source will be applied to the REF pin 6 of the instrumentation amplifier to level-shift the output so that the INA826 can drive a single-supply ADC. For this purpose, the REF pin 6 of the instrumentation amplifier will be connected to an LM741 operational amplifier configured as a voltage follower and powered by a ±12 V supply. The pin configuration of the LM741 is represented in Figure 7. Identify the different pins and build the circuit shown in Figure 8 by taking R= 100 kΩ.

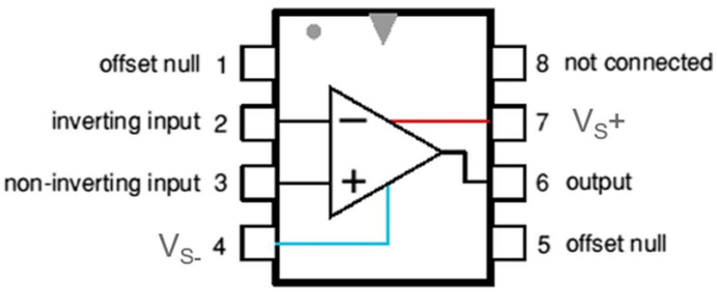

Figure 7. LM741 pin configuration

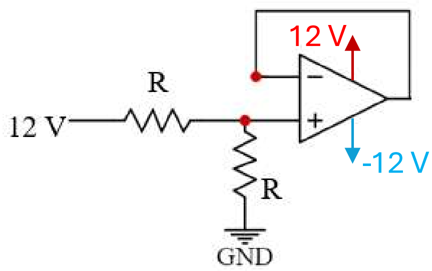

Figure 8. Voltage follower

Make the connection (Figure 9) between the instrumentation amplifier and the voltage follower by connecting the pin 6 (REF) of the instrumentation amplifier at the inverting input of the operational amplifier.

Please have the teacher check your circuit before switching on the voltage source.

- Measure the output voltage Vo of the instrumentation amplifier with no strain applied.

Place the different masses on the strain gauge and measure again the output voltage of the instrumentation amplifier.
Check if the entire signal range is strictly positive.
Give the range of the output values.

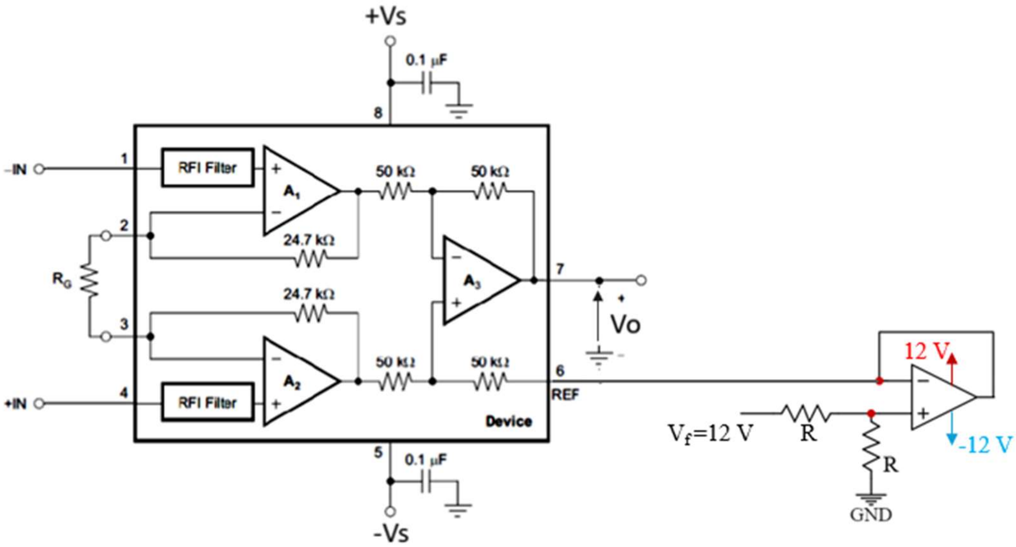

Figure 9. Instrumentation amplifier with voltage follower

- To make the signal compatible with Arduino, you must reduce the output voltage range to 0- 5 V.

Based on your previous results, modify your voltage follower circuit so that the output signal is scaled down to fit within the 0–5 V range. For this modification, a 5V power supply module for breadboard is provided. Justify your approach.

Please have the teacher check your circuit before switching on the voltage source.

Place the different masses on the strain gauge, measure again the output voltage of the instrumentation amplifier and check the output range.

- When no strain is applied, we want to obtain an output voltage of 2.50 V.

To achieve this, modify your circuit by adding a three terminal potentiometer P connected to the voltage follower circuit as shown below.

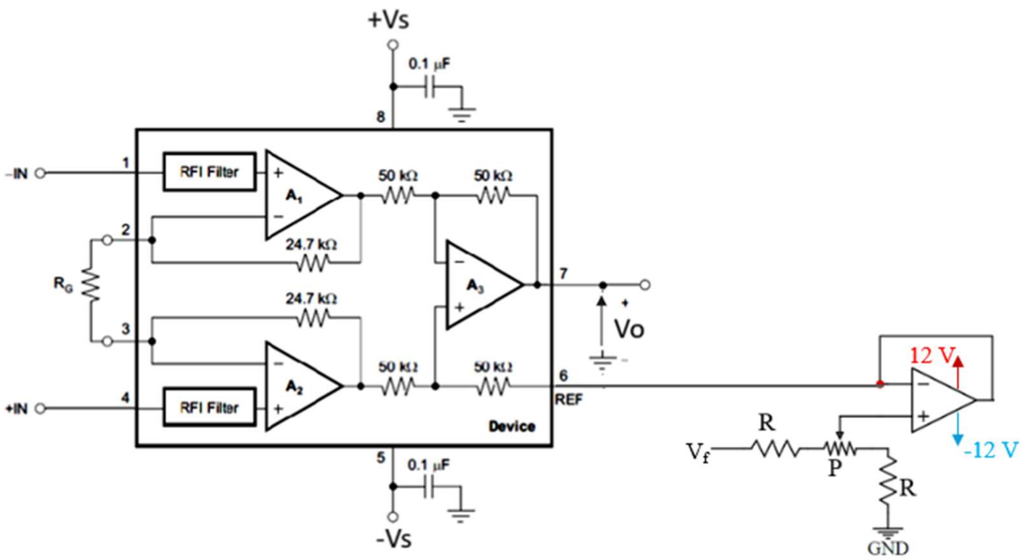

Figure 10. Instrumentation amplifier with voltage follower and potentiometer

- The position of the wiper is adjusted mechanically by using a screwdriver. Adjust the position of the potentiometer wiper so that the output voltage is 2.50 V when no mass is applied.
- Send the output voltage to Arduino and display the corresponding voltage in the serial monitor.
- Write a program that displays the corresponding mass based on the measured voltage
- Determine the maximum mass that can be measured by the system

5. Comparison of the different sensors

Compare the TAL220B strain gauge with your homemade sensor (Lab1) and the FlexiForce sensor (Lab1). Which sensor would you choose for a digital weighing scale?

Evaluation

The report should be submitted as a pdf file generated from the provided Word template on Moodle by April 26th, 2026, at 11.59pm.

This report must include the following sections: Introduction, Methods, Results, Discussion, and Conclusion.